

CAK2HAB3 - Dasar Kecerdasan Artifisial

Heuristic Search: Hill Climbing and Simulated Annealing

Lecturer Team





Outline

- ▶ Hill Climbing
- ▶ Simulated Annealing



Heuristic Search

- ▶ Hill Climbing
- ▶ Simulated Annealing (SA)
- ▶ Best-First Search
- ▶ Greedy Best-First Search
- ▶ A*



Heuristic Search

- ▶ Iterative Deepening A* (IDA*)
- ▶ Simplified Memory-Bounded A* (SMA*)
- ▶ Bi-directional A* (BDA*)
- ▶ Modified Bi-directional A* (MBDA*)
- ▶ Dynamic Weighting A* (DWA*)
- ▶ Beam A* (BA*)



Hill Climbing



Hill Climbing (HC)

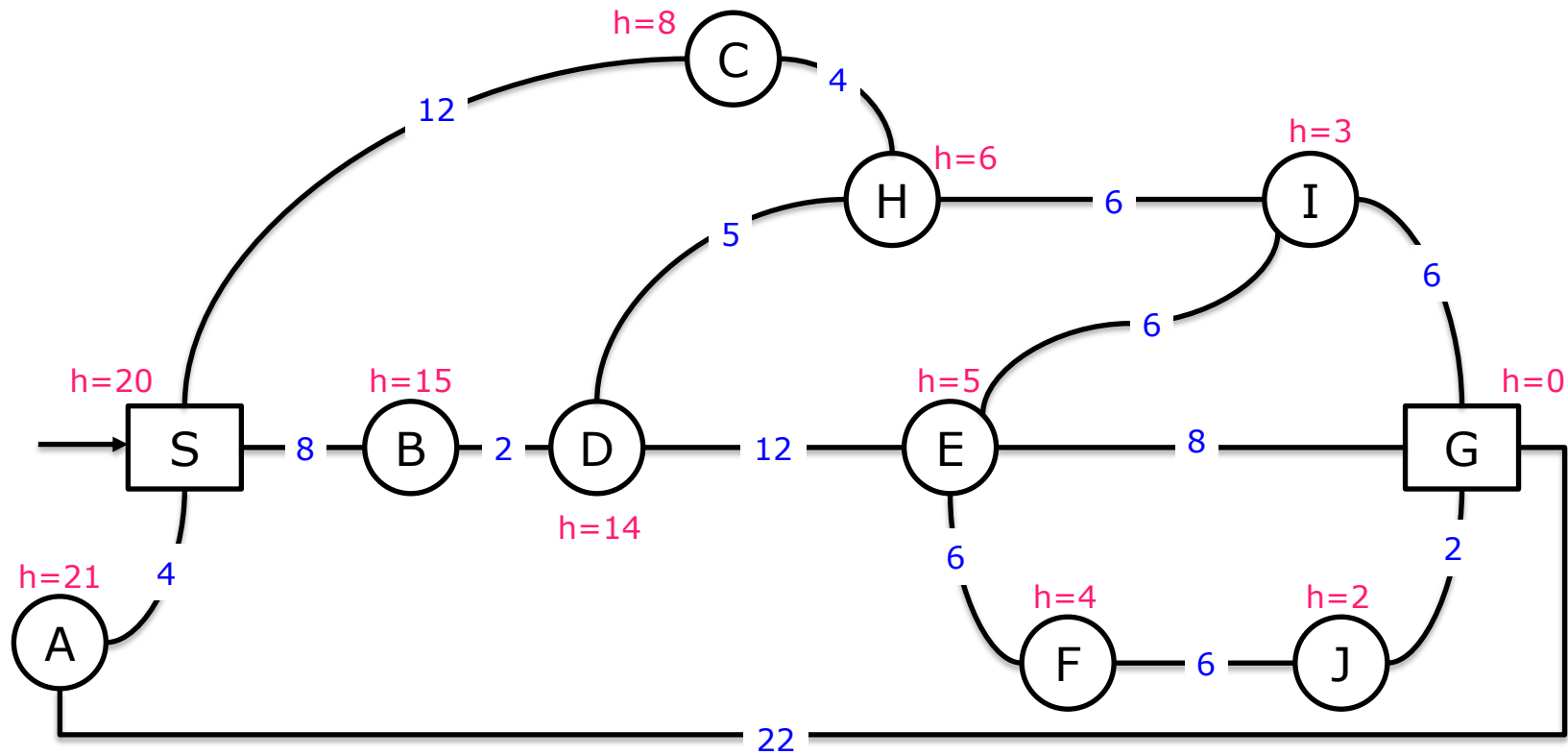




HC Algorithm

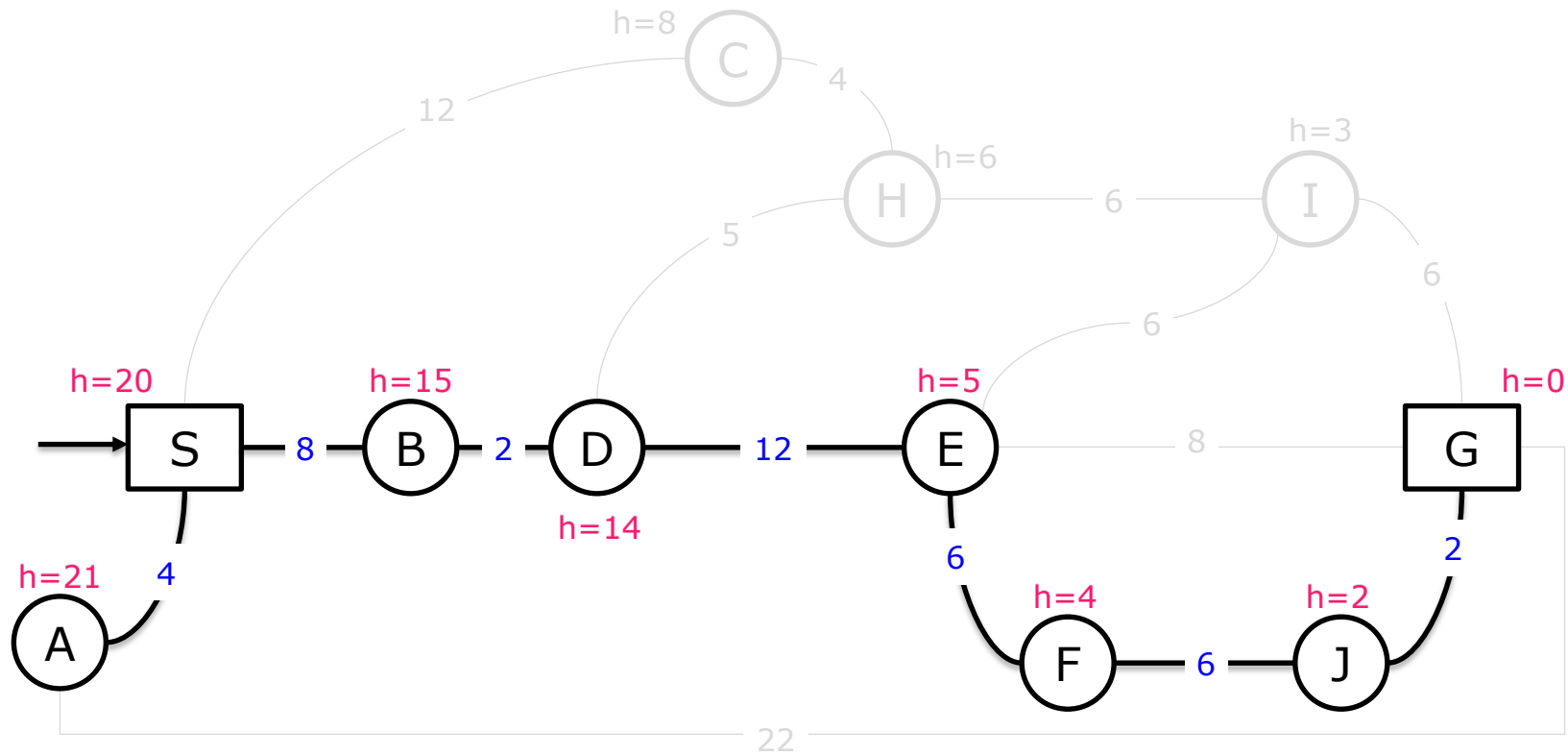
1. Evaluate the initial state if it is goal state quit otherwise current state is initial state.
2. Repeat until the current state is goal state or no new operators available:
 - Select a new operator for this state and generate a new state.
 - Evaluate the new state
 - if it is closer to goal state than current state, make it current state
 - if it is not better, ignore

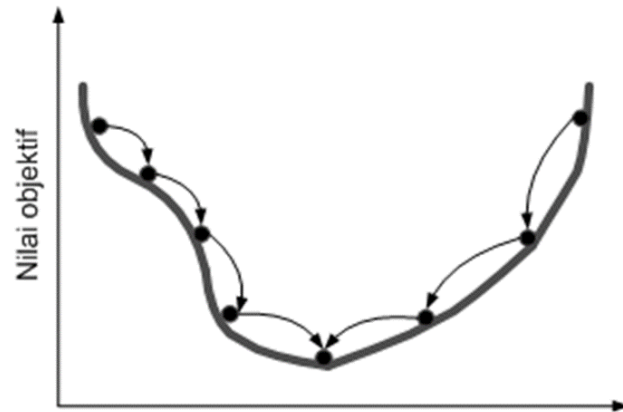
Problem Example



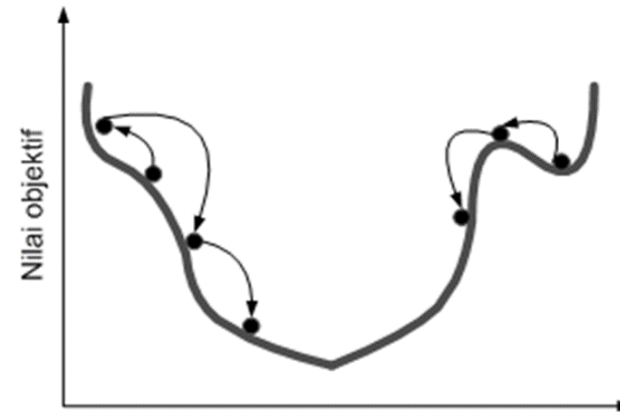


Simple Hill Climbing

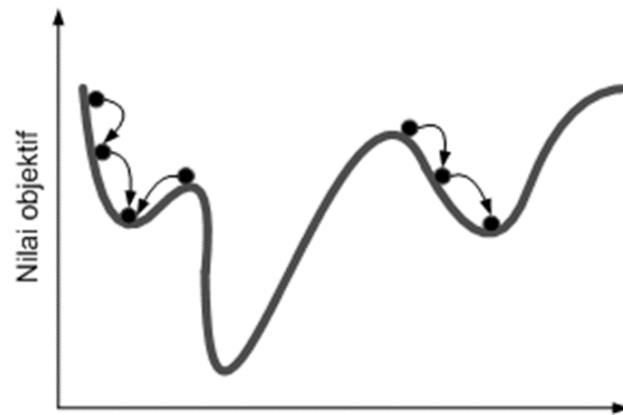




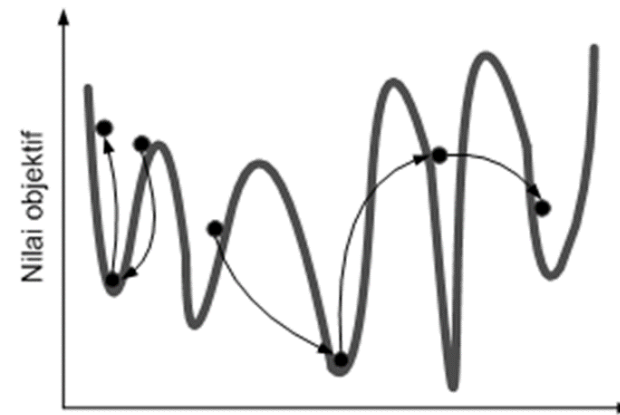
(a) *best case*



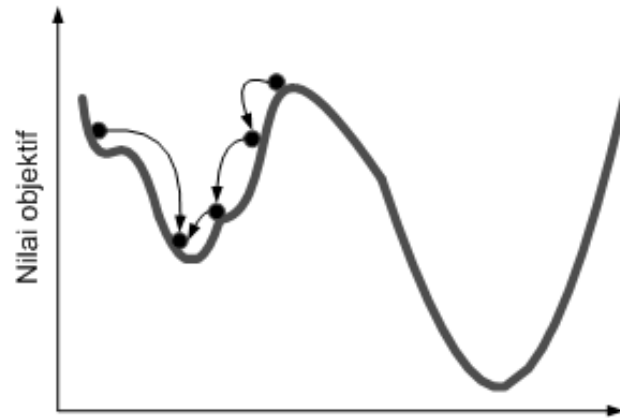
(b) *smooth*



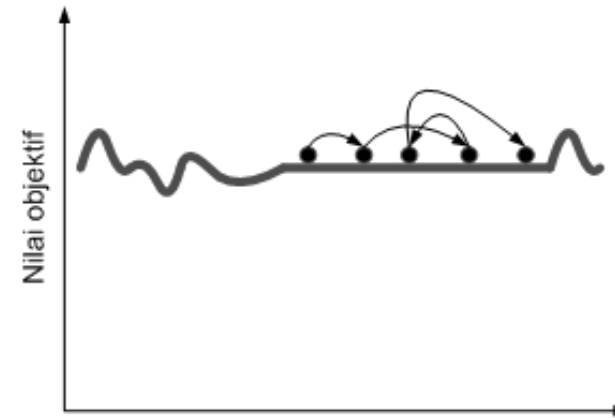
(c) *multimodal*



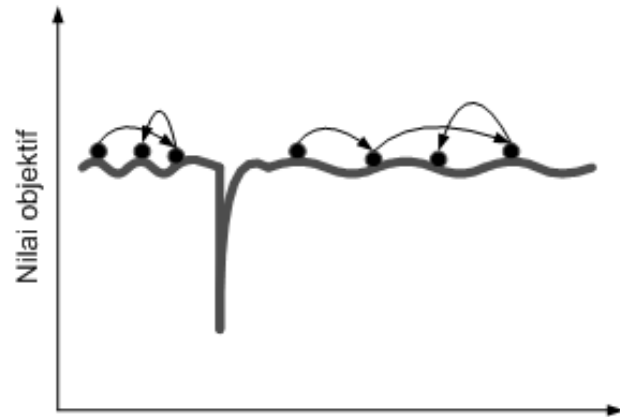
(d) *rugged*



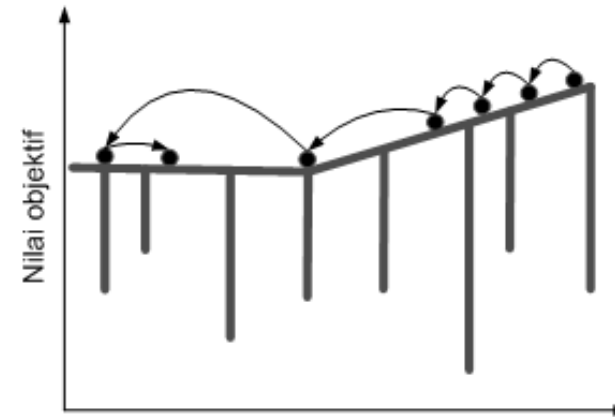
(e) *deceptive*



(f) *neutral*



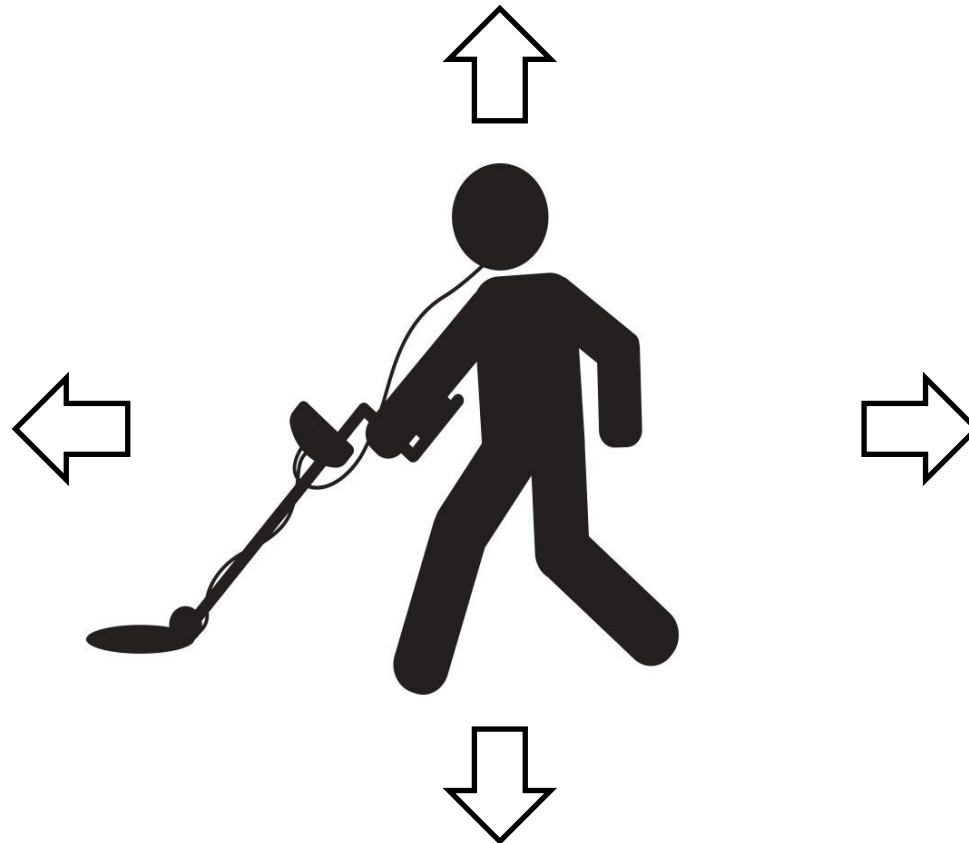
(g) *needle-in-a-haystack*



(h) *nightmare*



Metal Detector

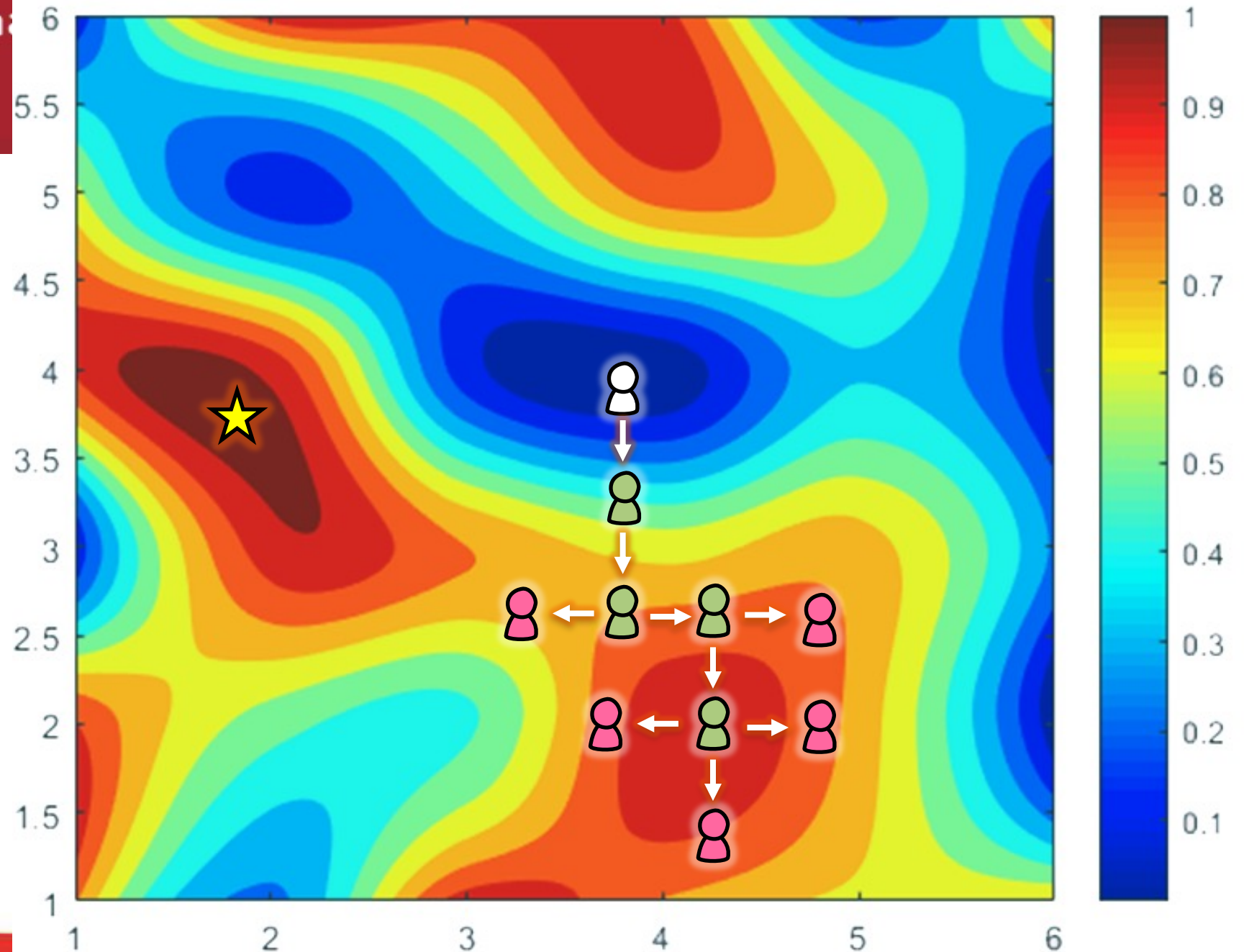


Operations:
1 step up, down, left, right



Metal Detector

Operations:
1 step up, down,
left, right





Steepest-Ascent Hill Climbing



Steepest-Ascent Hill Climbing

- In **simple hill climbing**, the first closer node is chosen, whereas
- In **steepest ascent hill climbing** all successors are compared and the closest to the solution is chosen
- This indicates that it has elements of the breadth first algorithm.

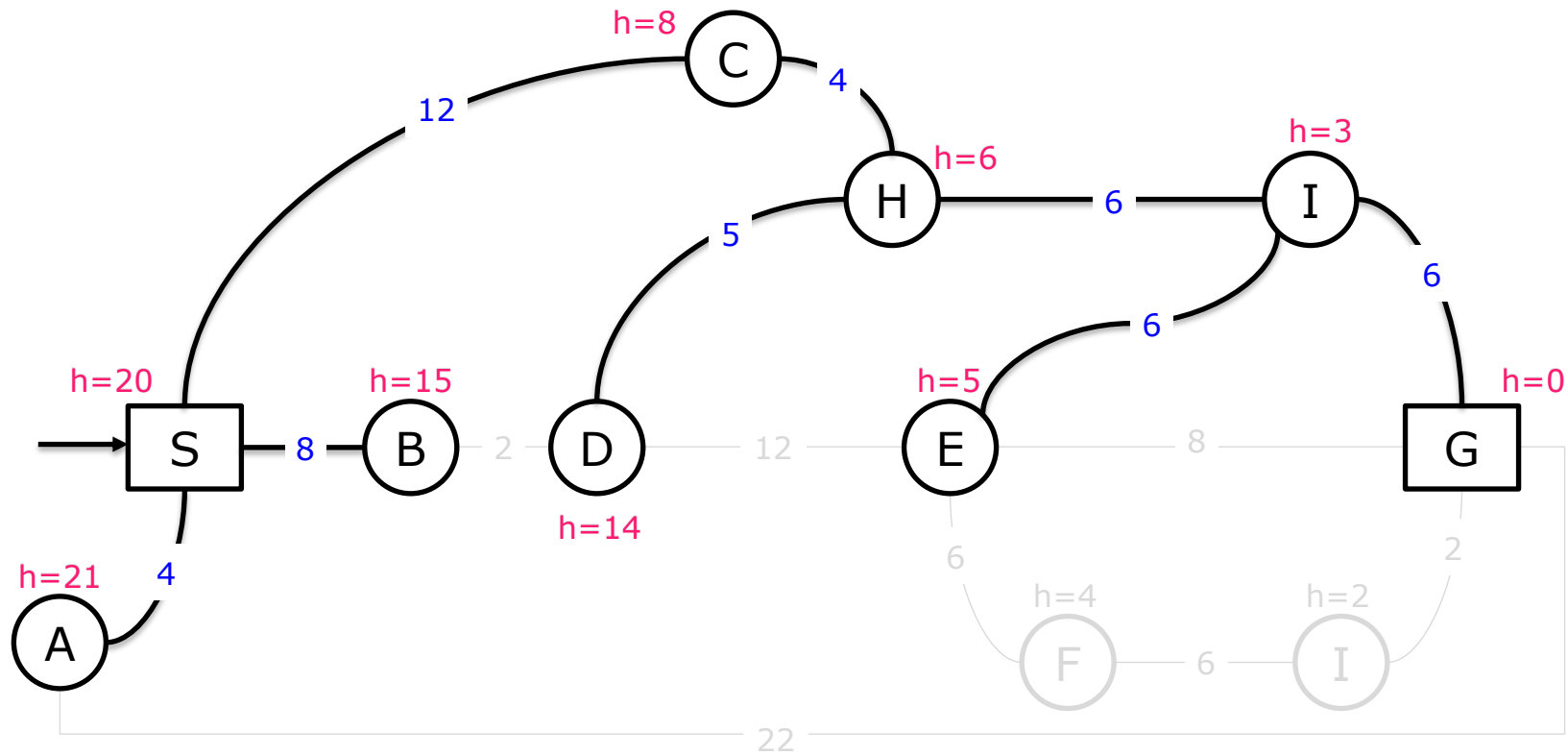


Steepest-Ascent HC Algorithm

1. Evaluate the initial state if it is goal state quit otherwise current state is initial state.
2. Repeat until the current state is goal state or current state does not change:
 - Save value of current state in a Target variable
 - For all operator available:
 - Generate and evaluate a new state
 - If new state is goal, set it as current state then quit
 - if new state is better than Target, save new state as Target
 - If Target is better than current state, set Target as current state



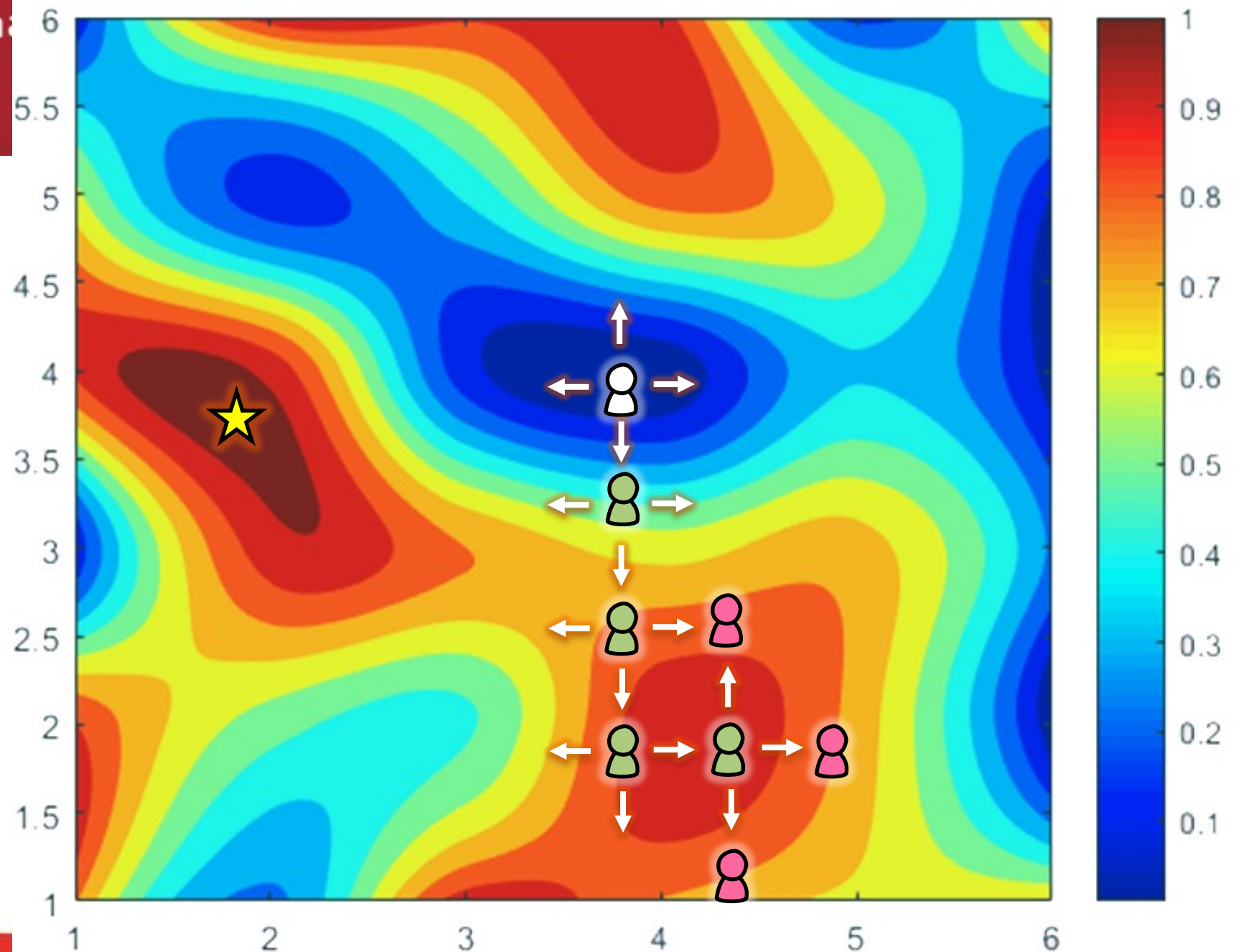
Steepest-Ascent Hill Climbing





Metal Detector

Operations:
1 step up, down,
left, right





Simulated Annealing



Simulated Annealing

- ▶ A variation on hill climbing and the idea is to include a general survey of the scene to avoid climbing false foot hills.
- ▶ The name and inspiration come from annealing in metallurgy, a technique involving heating and controlled cooling of a material to increase the size of its crystals and reduce their defects

Simulated Annealing

- ▶ SA uses a probability formula that allows it to get out of local minimum.
- ▶ New states that are no better than current states may still be selected with probabilities:

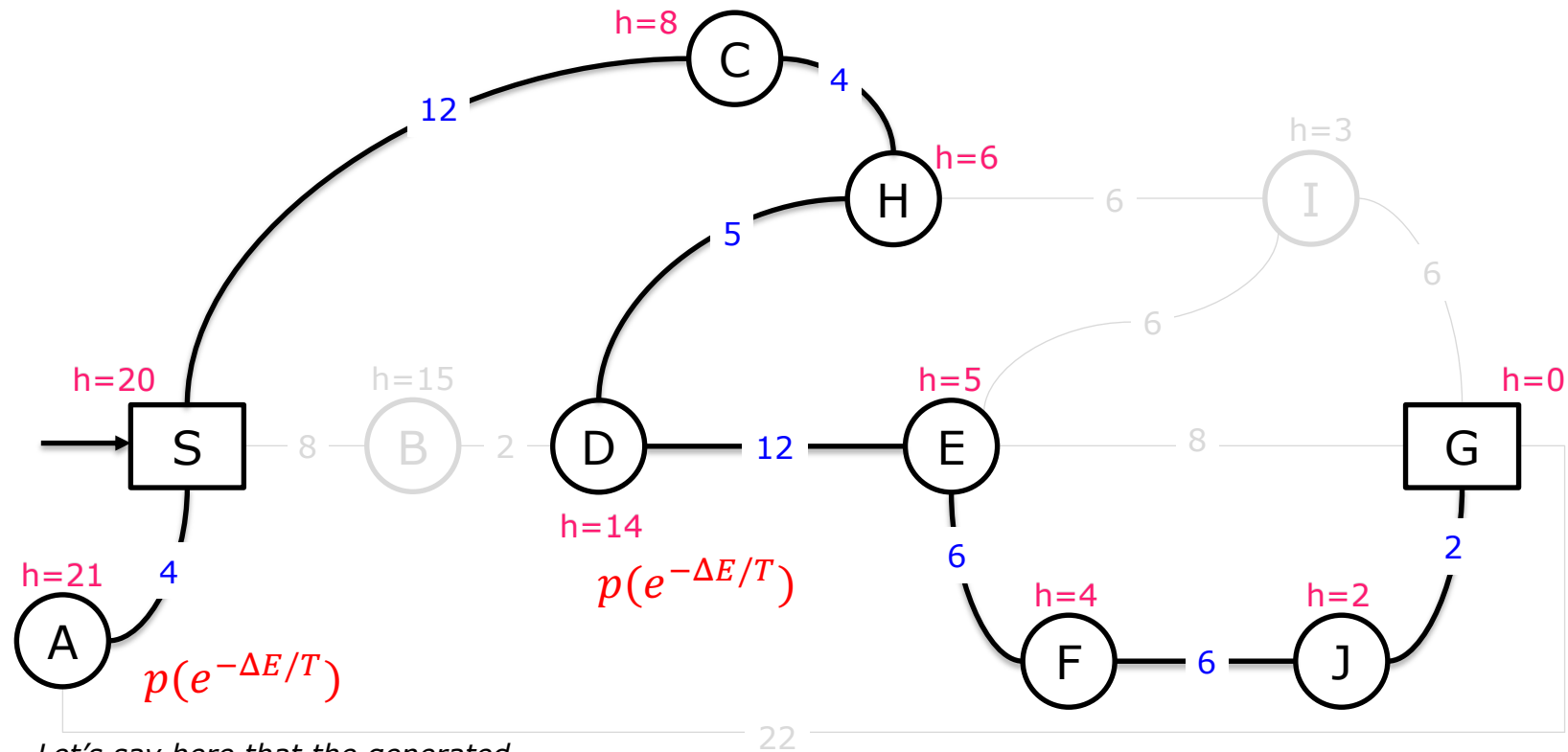
$$p(e^{-\Delta E/T})$$

- T is a control parameter called computational temperature, which controls the magnitude of the perturbations of the energy function $E(x)$.

Simulated Annealing Algorithm

- Evaluate the initial state if it is goal state quit otherwise current state is initial state
- Initialize T with a large value. (annealing schedule)
- Initialize Best-So-Far with current state
- Repeat until current state is goal or no new operators available
 - Select a new operator for this state and generate a new state
 - Evaluate the new state cost $\Delta E = f(\text{new state}) - f(\text{current state})$
 - If new state is goal or $\Delta E(x) < 0$, keep the new state as current state and Best-So-Far
 - If $\Delta E(x) \geq 0$, accept the new state as current state with probability
$$P = e^{-\Delta E/T}$$
 - Revise $T = T - \Delta T$ according to annealing schedule
- Return Best-So-Far as solution.

Simulated Annealing

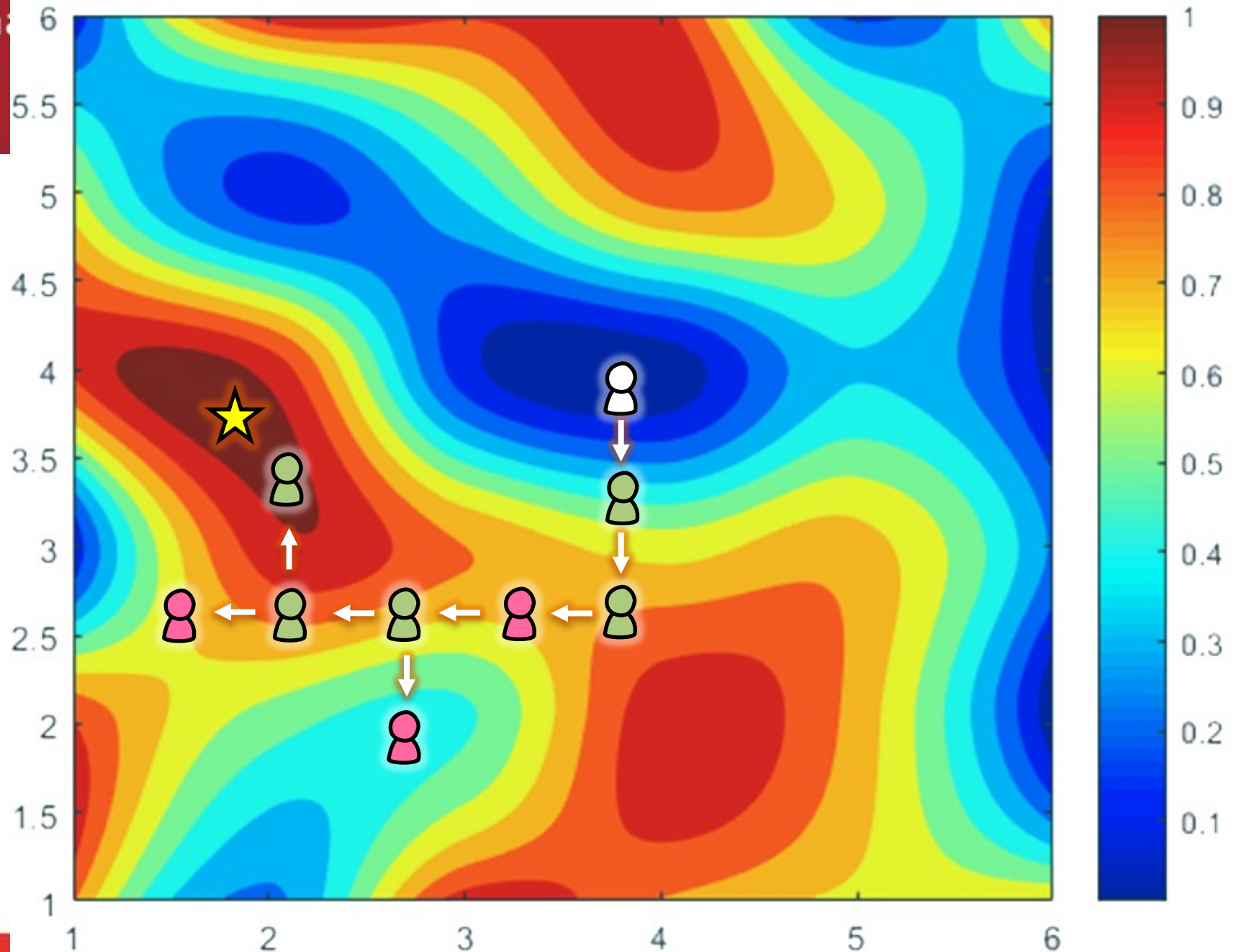




Metal Detector

Operations:
1 step up, down,
left, right

$$p(e^{-\Delta E/T})$$





Question?





THANK YOU